

A graph attention network framework for generalized-horizon multi-plant solar power generation forecasting using heterogeneous data

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ABSTRACT

Accurate forecasting of solar power output from multiple photovoltaic plants simultaneously for different time horizons is crucial for their large-scale integration into the electric grid. Forecasting strategies for PV output power significantly vary depending on the forecasting horizons, quality, variety, resolution of data, and fields of application. While the researchers addressed many of these particular cases to achieve high forecasting accuracy, the literature lacks sufficient discussion on integrating strategies for various forecasting scenarios into a general framework. This article proposes such a framework facilitating PV power forecasting with variable time horizons and imparting data of various types and granularity by introducing a single adjustable module into the framework. Moreover, by proposing a geographic distance-based graph construction, ensuring minimal vertex connectivity and adjustable sparsity, the technique captures the spatiotemporal correlation among the PV plant through a graph attention network for accurate forecasting. The proposed technique is highly scalable achieving excellent forecasting performance (an average absolute error of $< 5\%$ of the plant capacity for 5 min to 3-day forecasting horizon) up to thousands of PVs. The results indicate that the proposed method outperforms state-of-the-art approaches, such as Long Short-Term Memory networks, in terms of both accuracy and scalability. Additionally, this paper analyzes the suitability of features for forecasting in different scenarios and performance sensitivity to various model parameters.

1. Introduction

1.1. Motivation

Photovoltaic (PV) energy, also known as solar energy has emerged as a leading renewable energy source for its sustainability and environmental benefits. However, similar to other renewable energy sources, solar power generation outputs are inherently uncertain, intermittent, and contain high fluctuations due to meteorological conditions [1–3]. These variations are driven by solar irradiance influenced by factors such as the position of the sun due to the axial rotation and revolution of the earth, cloud cover, orientation, tilt, and materials of PV panels, and other atmospheric phenomena [1]. Since PV energy resources are being integrated heavily with the electric grids at present, it is of paramount importance to accurately forecast the output electrical power from the PV plants for the efficient operation, maintenance, and strategic planning of modern electrical grids ensuring grid stability, reliability, and maximization of carbon-free energy [3,4]. Effective forecasting supports critical functions such as resource allocation, electricity market operations, demand response programs, and the planning and development of grid infrastructure [4].

PV forecasting can be classified into long-term, medium-term, and short-term forecasting based on the *time horizon of forecasting*, defined as the amount of time between the present time (time instant at which the forecasting computations are performed) and the effective forecasting time (time instant for which the forecasting computations are performed) [2,3]. PV forecasting across different time horizons is crucial for addressing a range of issues related to power system operation, control, maintenance, and control [3,5]. Short-term forecasting applies to the maintenance and operational management of the power grid by assisting immediate adjustments and responses to fluctuations in power generation. Conversely, long-term forecasting is essential for strategic planning, as it informs decisions related to infrastructure development, investment, and policy-making [5].

Although solar power output is influenced by common weather factors, forecasting across different time horizons poses distinct challenges. These challenges arise from differences in the availability, characteristics, and quality of data, as well as the computational complexity of the models and the accuracy requirements for various grid-related applications. The literature presents a variety of methods and models tailored to specific forecasting time horizons, each necessitating unique

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types of data and information. However, there is a notable gap in the discussion of forecasting techniques adaptable to various forecasting horizons [6]. In other words, the literature lacks discussion on the integration of forecasting in different time horizons into a cohesive framework that accounts for their unique characteristics. To address this, we have developed a unified framework that facilitates forecasting across various time horizons by adjusting only a single module of the model, along with the requisite data and inputs. This integrated approach is designed to improve the ease, efficiency, and effectiveness of solar power forecasting across different temporal scales, thereby enhancing the overall utility and reliability of solar energy integration into the grid.

PV renewable energy plants are heavily integrated into the electric grid, contributing a considerable portion of the total power generation [7–9]. This is achieved by numerous small-capacity PV plants integrated into the grid. Assessing output power fluctuations in all these PV plants simultaneously is crucial for grid maintenance, stability, and avoiding stress on the grid [9,10]. Moreover, since the neighboring plants' output powers are correlated, capturing this correlation (usually through graph-based or similar structured data analysis) can aid in forecasting. While the literature typically deals with systems containing hundreds of PV plants, we extended this approach to include thousands of PV plants. An accurate prediction in a single PV site with detailed fluctuations can be crucial for local maintenance e.g., assessing stability and power quality issues at a few nearby buses, smoothing power output, and controlling ramp rate with [11,12], etc. However, simultaneous prediction in multiple sites, even with less accurate fluctuation tracking, provides the whole grid picture for renewable generation and facilitates situational awareness and decision-making in large power systems. Additionally, the same framework can be applied to smaller grids or parts of the grid with smaller graphs and provide more accurate fluctuation tracking with modification of a few parameters.

1.2. Related works

PV power forecasting has a well-established body of literature spanning several decades, encompassing a wide range of methodologies, and scenarios. Antonanzas et al.'s review work [2] provides a complete discussion of the basic concepts related to the literature of PV generation forecasting by introducing the basic considerations, parameters, and models of solar output prediction, various types of input data depending on the context and data-availability, different models for forecasting considering the horizon and resolution of forecasting, and context-dependent performance evaluation criteria of the forecasting models. A recent article, [6] provides an overview of the current state of research on PV power forecasting and outlines potential future research directions. Popular PV output power forecasting includes statistical time-series forecasting methods (e.g., ARIMA [13]), classical machine learning-based methods (e.g., random forest, neural networks, support vector machines, linear regression [14], hybrid KNN-SVM [15]), sequence prediction methods (e.g., LSTM [16], Bi-LSTM, GRU [17]) and advanced neural network-based methods (e.g., transformers [18], spatio-temporal attention [19,20], generative adversarial networks and auto-encoder [21]). The discussion in this section, however, would be limited to PV output power forecasting from multiple plants simultaneously capturing the spatio-temporal correlations through graph neural networks.

The application of graph neural networks to capture the spatial relationship among PV plants in multi-plant PV forecasting is relatively new. In recent years, a few works have shown the prospect of using GNN in systems with up to a few hundred PV plants. Simeunović et al. [22] proposed a temporal-spatial multi-window graph attention network for forecasting the output power of multi-site PV generators that considers the dynamic adjacency matrix in the GNN architecture obtained from the time-varying correlation among the output power of the PV sites. The proposed technique can achieve good accuracy

in 4 to 6 hours-ahead forecasting with many PV sites in the range of hundreds using weather data along with the power output data as the inputs. Verdone et al. [23] proposed a combination of GNN and a 1-D convolutional neural network to capture the site-to-site and temporal correlation, respectively, for forecasting output power in a multi-site scenario with explainability. Zhu et al. [24] proposed to consider the lead-lag correlation among the plants resulting from time differences of the airflow to build the dynamic adjacency matrices for GNN-based solar prediction. However, the system considered in this paper consists of only eight solar plants. Although, Cini et al. [25] forecast PV power in a system with 5016 plants with a 30-minute horizon, they do not consider any data other than the output PV power.

Apart from directly using GNNs, Li et al. [26] suggests a temporal convolutional network-based model to combine physics-based hourly forecasting and data-driven five-minute interval forecasting to obtain good performance in hours-ahead forecasting of solar power generation in a multi-site scenario. Additionally, this work identifies the most contributive neighboring site of the target site to facilitate information regarding cloud movement which is one of the most determining factors in precise forecasting of solar power generation. However, the proposed method utilizes data such as pyranometer-measured or reference cell-measured irradiance that is not always available in large-scale scenarios. Solar power generation scenarios can be very different depending on the location and the circumstances, and additional factors might be required to be considered when forecasting solar power output. For instance, Stenstadvolden et al. [27] showed the influence of snowfall on PV system generation output and suggested a multi-linear regression-based prediction model, especially for places with prolonged snowfall.

1.3. Research challenges and gaps

The main gap in multi-site PV output power forecasting literature lies in several aspects of the forecasting. The scalability of PV output power forecasting to systems containing thousands of PV plants is a major factor, essential for analyzing the effect of PV output prediction on large electric grids. Additionally, there is a need to understand the impact of each meteorological attribute on forecasting accuracy and their suitability for different time horizons in neural network-based forecasting for multi-site systems. Furthermore, developing a single framework that enables a wide range of forecasting horizons (from ultra-short-term to medium and long-term) with multi-site consideration remains a significant challenge [6].

1.4. Main contributions

Inspired by the aforementioned research gaps and challenges, we propose a GAT-based framework that is capable of predicting solar power output from multiple PV sites (a maximum of 1146 sites in six eastern states) simultaneously, which is very important from the perspective of grid operation, management, and planning. Moreover, the proposed GNN-based forecasting framework can be applied to short-time forecasting, as well as for increased time horizon forecasting along with day-ahead or multiple day-ahead forecasting. The selection of appropriate features for different time horizons can be accommodated within the same framework by only modifying one module of the GAT-based model.

The main contributions of the paper are as follows:

- This article proposes a single generalized graph attention network (GAT)-based framework for generalized-horizon solar power forecasting in a multi-site scenario relevant to the decision-making and planning of very large electric grids. The proposed framework is adaptive to forecasting horizons ranging from a few minutes to multiple days by adjusting a single module of the framework.

- The article introduces a geographic distance-based graph construction for GAT-based forecasting with adjustable sparsity while ensuring minimum connectivity of each vertex (PV plant). The proposed graph structure archives high accuracy in forecasting while ensuring the scalability of the proposed framework to very large systems with thousands of PV plants. Moreover, the proposed graph construction technique, being based on geographic distance, is robust to new types of meteorological features.
- This work analyzes and explains the contribution of different weather-related features depending on the time horizon of forecasting PV output enabling the selection of appropriate features according to the time horizon of prediction, required precision of forecasting, and availability, and granularity of data.

2. Method

This section provides the details of the proposed methodology of GAT-based forecasting by introducing a mathematical representation of the generalized-horizon solar power output forecasting from multiple PV plants simultaneously.

2.1. Mathematical representation

Let us denote the set of all PV plants (sites) in a multi-site PV system with P photovoltaic plants by $\mathcal{P}$; therefore, the cardinality of $\mathcal{P}$ is $|\mathcal{P}| = P$. The PV plants have different capacities of solar power generation; the maximum output power generation capacity of plant, $i \in \mathcal{P}$ is p_{cap_i} kilowatts (kW). The instantaneous output solar power generation by the i -th PV plants at a time, t is denoted by $p_i(t)$. Therefore, the capacity normalized output solar power of the i -th PV plant is: $x_i(t) = \frac{p_i(t)}{p_{cap_i}}$. The vector $\mathbf{x}(t) \in \mathbb{R}^P$ with elements $x_i(t)$ is the capacity normalized output solar power vector associated with the PV system at time, t . The forecast capacity normalized output solar power vector at any time, $t + t_h$ is expressed as:

$$\hat{\mathbf{x}}(t + t_h) = \mathcal{F}(\mathbf{x}(t), \mathcal{U}_{t+t_h}, \mathcal{M}_F), \quad (1)$$

where $t_h > 0$ is the forecasting horizon, $\mathcal{F}$ is the forecast operator, and $\mathcal{M}_F$ is the set of all parameters associated with the forecasting model. $\mathcal{U}_{t+t_h}$ is the set of information regarding the weather, date-time, miscellaneous information at the forecasting time instant, $t + t_h$, i.e.,

$$\mathcal{U}_{t+t_h} = \left\{ \hat{\mathbf{W}}(t + t_h), \mathbf{s}(t + t_h), d(t + t_h) \right\}. \quad (2)$$

In this article we consider, $\hat{\mathbf{W}}(t + t_h)$ is a $P \times W$ the forecast weather matrix containing W weather attributes (temperature, cloud cover, etc.) in each of its columns at the coordinate of each plant along each of the rows. $\mathbf{s}(t + t_h) \in \{0, 1\}^P$ is a binary column vector containing the sunset and sunrise time information at the coordinate of each plant along each of the rows. Any element, $s_i(t + t_h)$ of the vector, $\mathbf{s}(t + t_h)$ defined as: $s_i(t + t_h) = 1$, if $t_{i,sunrise} < t + t_h < t_{i,sunset}$, and 0, otherwise, where $t_{i,sunrise}$ and $t_{i,sunset}$ indicate the sunrise and sunset moments at the coordinate of the i -th plant. The scalar, $d(t + t_h)$ is the time-of-the-day index at time $t + t_h$ expressed mathematically as: $d(t + t_h) = \frac{rem(t, n_{Day})}{n_{Day}}$, where, n_{Day} is the number of temporal samples in a single day and $rem(a, b)$ indicates the remainder of the division $\frac{a}{b}$, where $a, b \in \mathbb{N}$.

2.2. Forecasting architecture

The conceptual architecture for implementing Eq. (1) to forecast the output power capacity normalized output solar power vector with a forecasting horizon of t_h at any time is illustrated in Fig. 1. The architecture consists of multiple modules: PV plant embedding, Graph construction module, input data processing module, the GAT module, and fully connected layers. The mathematical representations and operational details of those modules are described below.

2.2.1. PV plant embedding

The plant embedding module embeds all the plants into D -dimensional vectors, $\mathcal{E} : \mathcal{P} \rightarrow \mathbb{R}^D$. The embedding vectors, $\mathbf{v}_i, \forall i \in \mathcal{P}$ are higher-dimensional representations of the PV plants which are learned through the set of embedding weights, $\mathcal{E}_W$. The $D \times P$ embedding matrix, $\mathbf{V}$, contains the embedding vectors $\mathbf{v}_i$ in its columns in the same order as the plant numbers, $i \in \mathcal{P}$.

2.2.2. Graph construction module

Analyses of solar power generation from multiple PV sites spreading over a large geographical location under the GAT framework require defining the graph representation of the PV sites. The PV sites are in general considered as the vertices of the graph; however, the edges of the graph can be defined based on the geographic distance [23,25] as in static graph or using the dynamic correlation among the output of the PV sites [22]. Using dynamic graphs has certain advantages in capturing the spatiotemporal relationship between the output power of different sites, however, in this work, a geographic distance-based graph is preferred due to several reasons. Firstly, in a forecasting scenario with a large number of PV sites, a fixed graph would reduce the computational cost of the GNN-based data-driven methods. Secondly, a variety of data has been proposed to utilize as the input of the prediction model depending on the forecasting horizon length, h , and the availability of the data. Consequently, using a static graph facilitates including new types of data, as the forecasting input, with minimal change in the model architecture. Since most of the data proposed to be used and considered usable by the researchers depending on availability and applicability are related to weather and geographic aspects; therefore, a geographic distance-based static graph should be a more generalized choice to accommodate heterogeneous input data. Fig. 2(a) illustrates the average total cloud cover values in 2006 at the location of the PV sites within our area of consideration. The smooth change of cloud cover values across geographic distances justifies the consideration of a geographic distance-based graph.

Fig. 2(b) also shows that due to the patterns of the PV site locations within the area of consideration, a direct geographic distance threshold-based graph forming similar to [23] may be challenging. Specifically, the large variation among the geographic distance values between the pairs of the PV sites poses a trade-off between too many edges for some of the vertices and disconnected graphs. Therefore, we propose the following technique of forming the graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$ for GAT-based forecasting, where $\mathcal{V} = \{v_1, v_2, \dots, v_P\}$ is the set of vertices of the graph corresponding to the PV sites, $\mathcal{E}$ and $\mathcal{W}$ are the sets of edges and their corresponding weights, respectively, $P = |\mathcal{V}| = |\mathcal{P}|$ being the total number of vertices, PV plants in the area of consideration.

For the construction of the graph let us define a binary matrix, $\mathbf{A}_{temp}$ with elements $a_{Temp_{ij}}$ as:

$$a_{Temp_{ij}} = \begin{cases} 1, & \text{if } d_{ij} \leq d_t \text{ or } v_j \in \mathcal{N}_i^{(K)} \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where d_{ij} be the geographical arc length distance of the between i -th and j -th PV sites calculated from their latitudes and longitudes, and $\mathcal{N}_i^{(K)}$ is the set of K -nearest neighbors of v_i based on the distance, d_{ij} . The second criterion ($v_j \in \mathcal{N}_i^{(K)}$) in Eq. (3) for the existence of an edge e_{ij} ensures that any plant represented by $v_j \in \mathcal{V}$, which has no other plant in its vicinity (within distance d_t), would still be connected to the graph. However, the degree of similarity in output power fluctuations between that plant, v_j , and the plant(s) to which it is connected would be automatically determined by the associated attention scores learned through the training process. Since the matrix, $\mathbf{A}_{temp}$ is asymmetric, a symmetric version of $\mathbf{A}_{temp}$ defined as: $\mathbf{A} = \mathcal{U}(\mathbf{A}_{temp})$, is considered as the final graph adjacency matrix. Here, $\mathcal{U}$ is a matrix operator that replaces the lower triangular portion of the matrix with its upper triangular portion to produce a symmetric matrix.

The parameters, d_t and K are related to the sparsity of the graph and can be tuned depending on the application, desired level of forecasting

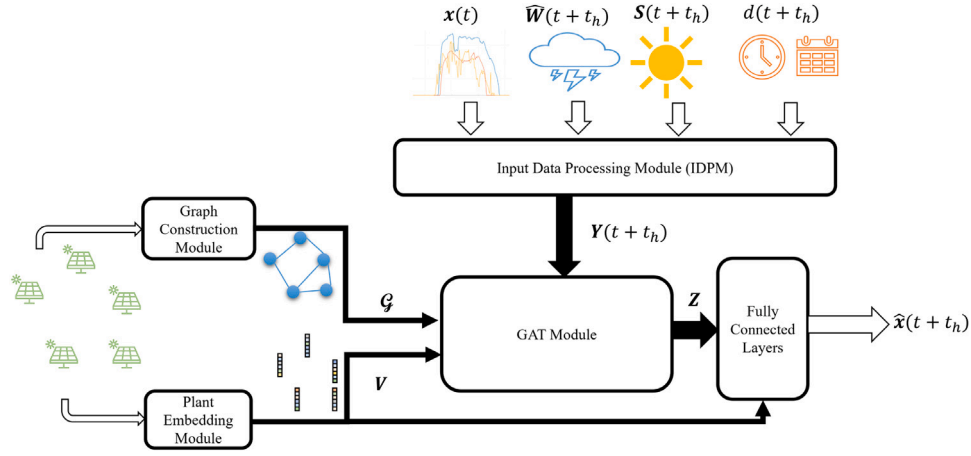
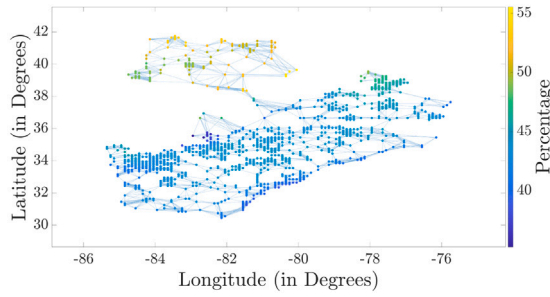
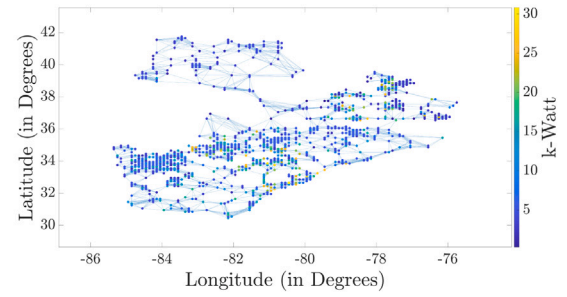


Fig. 1. Schematic diagram of the proposed framework.



(a) Average total cloud coverage over a year, 2006, as a graph signal.



(b) Average total PV output over the year, 2006, as a graph signal.

Fig. 2. Multi-site PV system as a graph.

precision, and computational resources. This tunable graph sparsity makes the proposed framework robust to various types of grid applications. The binary adjacency matrix, A (with elements a_{ij}) can be converted to a weighted adjacency matrix, W with elements w_{ij} as:

$$w_{ij} = \begin{cases} \mathcal{S}(\frac{1}{a_{ij}}), & \text{if } a_{ij} = 1 \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

where $\mathcal{S}$ is the appropriate scaling operator. $w_{ij} \in \mathcal{W}$, $e_{ij} \in \mathcal{E}$, if $w_{ij} \neq 0$.

2.2.3. Input data processing module

The input data processing module (IDPM) maps the heterogeneous input data to a fixed vector (i.e., node features) corresponding to each vertex (PV plants) of the graphs for forecasting at the time instant $t + t_h$. Let L be the length of the fixed vector, $y_i(t + t_h)$ corresponding to each vertex, $i \in \mathcal{V}$ of the graph constructing the IDPM output matrix, $Y(t + t_h)$ of size $L \times P$. The heterogeneity of the input data results from including different types of meteorological data and other information in the forecasting framework. In other words, the availability of data corresponding to different weather attributes, the time and spatial resolution of the weather data, and the inclusion of the date and time information contribute to the heterogeneity of the data. The IDPM resamples, performs mathematical processing, and concatenates various types of data to integrate them in a fixed-size matrix, enhancing the flexibility and robustness of the framework under multiple scenarios and data availability. Moreover, the IDPM can contain data processing involving learnable parameters facilitating neural-network-based data pre-processing. Here are some examples of implementation of the IDPM

in solar output power forecasting used in this work and usable in the extension of the work:

1. For short forecasting horizons, where only the output power data is sufficient for forecasting, all the L elements of the vector $y_i(t + t_h)$ are previous samples, $x_i(\tau)$, $\tau < t$ of output power data corresponding to each plant.
2. For longer horizon forecasting, where inclusion of the available weather attributes and day-time information improves forecasting performance, W weather attributes, one value related to sunrise and sunset, and one time of the day index are concatenated to $L - W - 2$ elements of previous samples of output PV power to obtain the vector, $y_i(t + t_h)$ for each plant.
3. In case of diverse temporal granularity of data, linear interpolation is used to up-sample the lower resolution data. However, in some cases, linear or state-of-the-art interpolations are not suitable for certain types of weather attribute data (e.g., the total cloud cover data contain high uncertainty) and may lead to lower accuracy. In those cases, if the scenario permits, down-sampling the higher-resolution data may provide better forecasting accuracy, although at a lower temporal granularity.
4. In day-ahead, or multiple day-ahead predictions, previous output power samples corresponding to or around a particular time of the day may be included as elements of $y_i(t + t_h)$.
5. Simple mathematical operations on temporal data (e.g., moving average) can be performed.
6. IDPM can adjust the mismatched dimensions and missing values within the data set through small mathematical operations such as concatenation, zero padding, or moving average before sending the data stream to training and testing.

7. Machine learning and neural network models with learnable parameters can be used to capture the inter-feature correlations.
8. To better capture the temporal correlation long-short-term-memory (LSTM), or other sequence processing techniques with learnable parameters can be used in the IDPM.

Although many of the techniques discussed above are popular data pre-processing techniques, utilizing them within the IDPM offers flexibility and adjustability of the proposed framework for implementation under various scenarios considering scalability, data availability, and application pragmatics.

2.2.4. The GAT module

The plant embedding matrix, $\mathbf{V}$, the graph, $\mathcal{G}$, and output of the IDPM, $\mathbf{Y}(t + t_h)$ are the inputs of the GAT module, while the matrix, $\mathbf{Z}$ containing the columns of $\mathbf{z}_i$ is the output. This module consists of a linear operation followed by a graph of attention networks to calculate the attention scores and use the score to calculate the GAT module output. The linear operation was implemented by the matrix multiplication of the IDPM output matrix, $\mathbf{Y}(t + t_h)$, with the $D \times L$ learnable weight parameter matrix, $\mathbf{L}_w$. The resultant matrix is subsequently concatenated with the embedding vectors to create the gate matrix, $\mathbf{G}$, of graph attention network [28,29]:

$$\mathbf{G} = \begin{bmatrix} \mathbf{V} \\ \mathbf{L}_w \mathbf{Y}(t + t_h) \end{bmatrix}.$$

The query ($\mathbf{q}$) and the key ($\mathbf{\kappa}$) of the attention network are calculated from the gate matrix, $\mathbf{G}$, respectively, as:

$$\mathbf{q} = \mathbf{a}_q^T \mathbf{G},$$

$$\mathbf{\kappa} = \mathbf{a}_\kappa^T \mathbf{G},$$

where $\mathbf{a}_q$ and $\mathbf{a}_\kappa$ are both $2D$ -dimensional vectors with learnable parameters as entries and $\begin{bmatrix} \mathbf{a}_q^T \\ \mathbf{a}_\kappa^T \end{bmatrix}$ is called the *attention weight matrix*. The graph attention scores are calculated as:

$$\alpha_{ij} = \text{SOFTMAX}_i \left(\text{LeakyReLU} \left((\mathbf{Q} + \mathbf{K}^T) \otimes \mathbf{A} \right) \right),$$

where $\mathbf{Q}$ and $\mathbf{K}$ are two $P \times P$ matrices having $\mathbf{q}$, and $\mathbf{\kappa}$ in each of the columns, respectively, and $\otimes$ denote the Hadamard product, i.e., element-wise matrix multiplication. LeakyReLU is the non-linear activation function operator [30] and SOFTMAX_i [31] signifies applying the Softmax operator along the i -th row of the matrix. The attention scores, α_{ij} , are then used to obtain the aggregated representation of the i -th vertex as:

$$\mathbf{z}_i = \text{ReLU} \left(\sum_{j \in \mathcal{N} \cup \mathcal{B}_1(i)} \alpha_{ij} \mathbf{L}_w \mathbf{Y} \right),$$

where $\mathcal{N} \cup \mathcal{B}_1(i)$ is the first hop neighboring vertices of vertex n . The matrix, $\mathbf{Z}$, contains the $\mathbf{z}_k$, in its columns which passes the fully connected layer as inputs. A detailed description on the GAT can be found in [28].

2.2.5. Fully connected layer

The final forecast vector is then obtained through a fully connected neural network, expressed as an operator, $\mathcal{P}$, with a set of learnable parameters, F_w , according to the following equation:

$$\hat{\mathbf{x}}(t + t_h) = \mathcal{P} [\mathbf{Z}^T \mathbf{V}].$$

3. Analysis of PV and weather data and simulation details

3.1. Data-set description

We evaluate the performance of the proposed GAT-based technique in predicting solar power output from a large number of PV sites (up to 1146 sites). The dataset for the performance evaluation is formed with renewable energy output data, location and types of the PV sites, weather data, sunset and sunlight times, etc described as below.

3.1.1. Solar power output data

The solar power generation data is from the *Solar Power Data for Integration Studies* [32] dataset of National Renewable Energy Laboratory (NREL). The data set is built synthetically using real weather data from the year 2006 at the locations of the PV plants in all 50 US states using the *Sub-Hour Irradiance Algorithm* [33]. In this study, data from 1146 PV sites in total from six eastern states: Ohio, West Virginia, Virginia, North Carolina, South Carolina, and Georgia are used. The data set provides solar power generation in megawatts (MW) for each PV site at a temporal resolution of 5 min. Besides, the dataset provides the geographical coordinates of each PV site (as longitude and latitudes), the capacity of the PV site in MW, and type of the PV site, i.e., utility or distributed PV sites [32].

3.1.2. Weather data

Weather data of the year 2006 have been collected from *National Oceanic and Atmospheric Administration, PSL Climate Data Repository* [34] with a temporal resolution of 3 h and spatial resolution of 277 and 349 points respectively for longitude and latitude. The weather attributes at the coordinates of the PV sites are interpolated using linear interpolation. The considered weather attributes are temperature, total cloud covering, albedo, and solar parameter canopy conductance. More weather attributes can be integrated into the framework through the IDPM depending upon the availability and application.

3.1.3. Sunrise and sunset data and the time of the day data

The exact time for sunrise and sunset at the place of a PV site has been calculated from its longitude and latitude using the technique described in [35]. The information are integrated via the IDPM through the vector, $\mathbf{s}(t + t_h)$, as described in Section 2.1. Apart from the types of data described above, the time-of-the-day (ToD) index, denoted as $d(t + t_h)$ as defined in Section 2.1 is crucial for forecasting with horizons from 1-hour to several days. Moreover, in case of the availability of multiple-year data, the information about the days of the year can be useful to enhance forecasting performance.

3.2. Relative importance among the features

In this work, extensive experiments on the NREL data set [32] reveal that the choice of features according to the forecasting horizon, especially the weather-related features, can enhance the accuracy of forecasting significantly. Automatic feature selection techniques by complex machine learning methods have proved to be effective in various forecasting applications. However, in this particular case of PV forecasting with a generalized time horizon, proper domain knowledge-inspired features selection and their appropriate processing by the IDPM increases the efficiency and scalability of the framework. Here, the relative importance of the features for forecasting in time various horizons are analyzed through visual inspection of the time-series power output and weather attribute data (Fig. 3), the column sum of the linear parameter matrix, $\mathbf{L}_w$ (Fig. 4), and finally by forecasting accuracy with different feature sets (Fig. 5).

The PV output power at a particular PV plant and the temperature, total cloud cover, and solar parameter canopy conductance at the coordinate of the plant corresponding to three different days in three seasons of the year have been illustrated in Fig. 3(a)–(d). From the within-a-day and across-days variation, it can be observed that the PV output power increases with the increase of temperature and solar parameter canopy conductance, whereas decreases with the increase of total cloud cover. Since the actual time resolution of the weather attribute time series (b)–(d) is 3 h and is interpolated using linear interpolation within the 3-hour interval, the fluctuation of PV output power within this 3-hr time period cannot be forecast solely by the weather attribute data of such temporal resolution. Therefore, for forecasting in a time horizon of less than 3-hr, the inclusion of the available weather attribute may decrease the forecasting accuracy. This

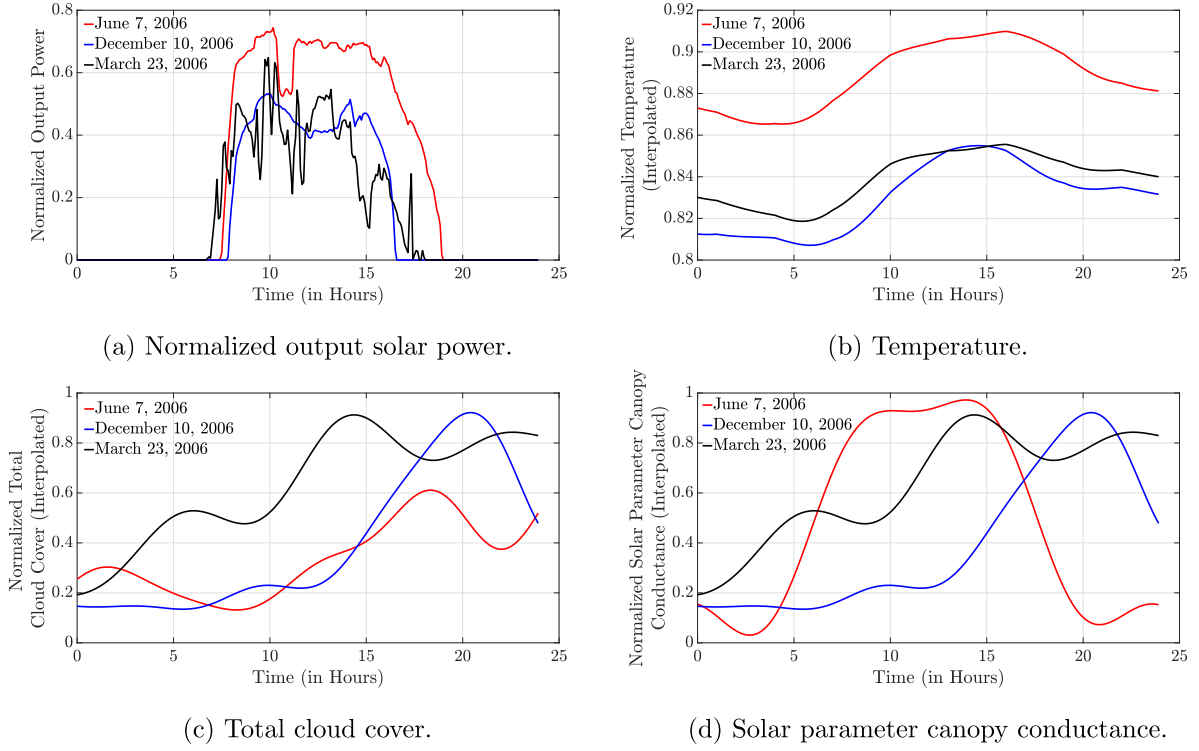


Fig. 3. Example day-long data: Analysis of the dependency of output PV power on the weather attributes.

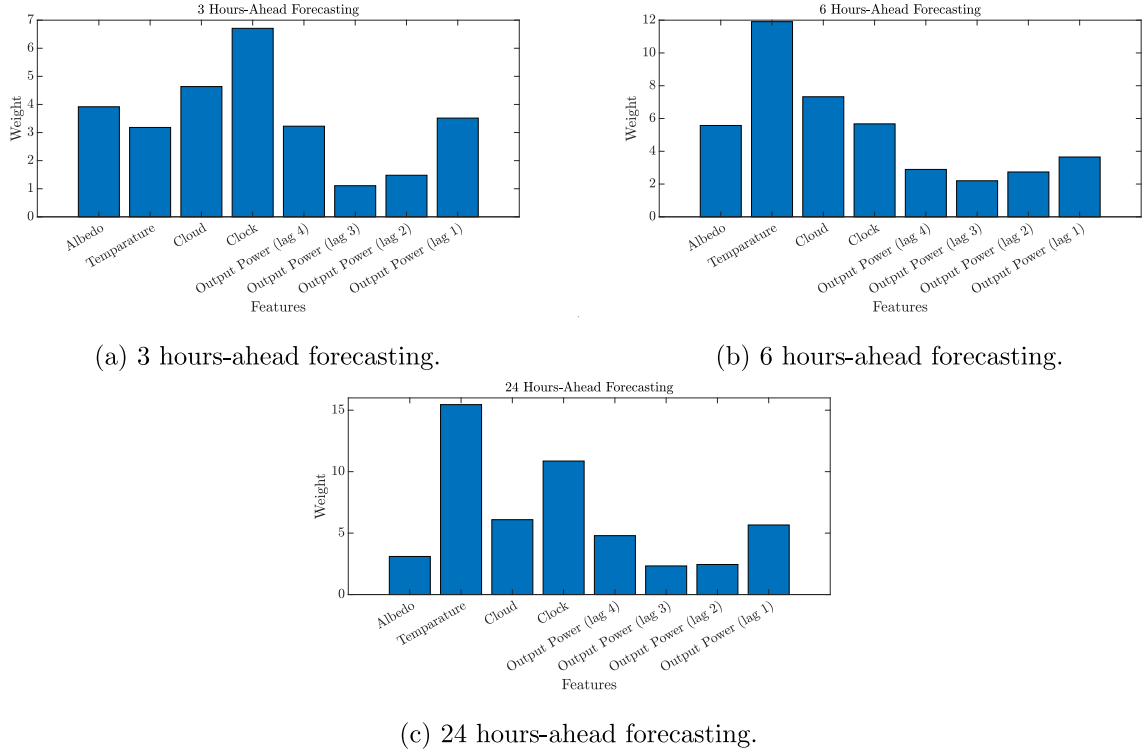


Fig. 4. Relative importance of various features in terms of the elements of the vector, $\mathbf{1}^T \mathbf{L}_w$.

is because the quality of weather forecasting, which involves spatiotemporal interpolation, may not be reliable, leading to large errors. In this case, forecasting with previous PV output power samples provides the best forecasting accuracy. On the other hand, our experiments show that in 1 to 3 h ahead forecasting inclusion of the time of the day index, $d(t + t_h)$ can improve the accuracy to some extent.

Fig. 4 presents the absolute values of the elements of the vector, $\mathbf{1}^T \mathbf{L}_w$, after training. Since, the matrix, $\mathbf{L}_w$ contains the weights associated with L elements (that may contain the past output values, weather-related features, and time-of-the-day depending on t_h) of the IDPM output vector $\mathbf{y}_i(t + t_h)$ along each embedding vector dimension, The row sum, $\mathbf{1}^T \mathbf{L}_w$, represents the relative importance of the L

Table 1

Summary of discussion on feature selection: sets of features in various range of forecasting horizons..

Range of forecasting horizon, t_h	Important features	Feature used in the experiments in this article
5 min–30 min	Only past output power samples	8 output power samples
30 min–6 h	Past output power samples and weather features (forecast) at the moment of forecasting	4 output power samples, 1 sample each for time-of the day, total cloud cover, temperature and solar power canopy conductance.
Days-Ahead	Mainly weather features	2 output power samples, 1 sample each for time-of the day, total cloud cover, temperature and solar power canopy conductance.

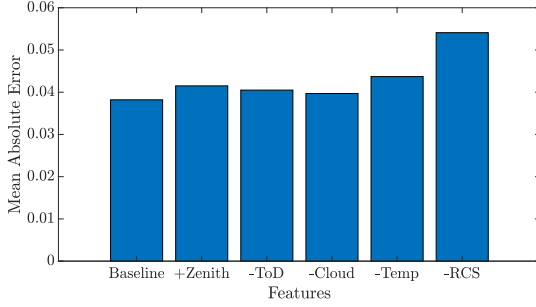


Fig. 5. Day-ahead forecasting performance with different sets of weather attributes. The baseline set consists of time of the day (ToD), temperature (Temp), total cloud cover (Cloud), and solar parameter canopy conductance (RCS) readings.

features. Fig. 4(a) justifies the inclusion of the time-of-the-day index, $d(t + t_h)$ for 3-hr horizon. Fig. 4(a)–(c) reveals that the importance of the past few output power samples around the forecasting moment, t decreases with the time horizon of forecasting t_h . This suggests that accurate forecasting over longer horizons is challenging when relying solely on the output power time series. Incorporating additional information, such as weather data related to the forecasting moment is necessary. Moreover, the time-of-the-day index, $d(t + t_h)$ is important for longer t_h . Among the weather attributes, temperature appears to be a more stable predictor than the total cloud cover or albedo. This is because the total cloud cover or albedo contains high fluctuations that cannot be captured by linear temporal interpolation, whereas the temperature changes are relatively slower.

Finally, Fig. 5 illustrates the day-ahead forecasting accuracy in terms of mean absolute error with reduced 3-hr resolution PV output power data to synchronize with the resolution of the weather attribute data. The baseline set consists of the time-of-the-day (ToD), temperature (Temp), total cloud cover (Cloud), and solar parameter canopy conductance (RCS) data along with two previous power output samples around the forecasting moment, t . The results show that the inclusion of the Zenith angle does not improve the forecasting accuracy and the exclusion of solar parameter canopy conductance introduces more forecasting error. It is noted that, for all these cases, sunrise and sunset times are included through the vector $s(t + t_h)$ and directly used during forecasting (inference) without including them as the input feature for training. A summary of the discussion has been provided in Table 1.

3.3. Parameter setting

Apart from the IDPM, the parameters in the other modules are kept constant for all the cases which enables the model to be implemented in different scenarios by only adjusting the IDPM. In the model configuration, the number of epochs is set to 100, with a mini-batch size of 32 and a learning rate of 0.001. For the hyper-parameters, the plant embedding vector dimension, D , is specified as 10, while the fully connected module consists of two layers with 256 neurons each.

4. Performance analysis

In this section, the performance of the proposed framework has been analyzed for forecasting PV power output for all plants simultaneously. For analyzing the forecasting performance for various time horizons, the sets of features and the processing technique in the IDPM have been selected according to the discussion in the previous section. For shorter horizons, using only previous output power samples yields the best results. The recommended range of horizon for using only previous output power samples depends on the time resolution of forecasting and the available resolution of weather data; however, for the available data set, it is selected as 5–30 min. For 1–3 h forecasting horizons, the time-of-the-day (ToD) indices are concatenated with the previous output power samples to provide the best results. For forecasts exceeding 3 h, a combination of previous output power samples, time-of-the-day (ToD), and weather attributes are considered.

For the analysis of the performance of the proposed framework and comparison with the benchmark techniques, the forecasting errors corresponding to every plant, $i \in \mathcal{P}$ at every test time instant, $t + t_h \in \mathcal{T}_{Test}$ is calculated as: $\hat{e}_i(t + t_h) = \hat{x}_i(t + t_h) - x_i(t + t_h)$. The mean absolute error is considered as the primary metric for performance evaluation in all the cases, which is defined as:

$$MAE = \frac{\sum_{i \in \mathcal{P}} \sum_{t+t_h \in \mathcal{T}_{Test}} \hat{e}_i(t + t_h)}{P |\mathcal{T}_{Test}|}. \quad (5)$$

Since the MAE is calculated as the average absolute forecasting error of capacity normalized PV power output, it is automatically plant-independent and always provides a normalized measurement of errors being bounded in the range $0 \leq MAE \leq 1$. However, relative error measurements are considered in some cases to provide a complete picture of error analysis. In the PV power forecasting scenario, the mean relative errors (MRE) are severely affected by the extremely high relative error values; due to many near-zero values of actual output power associated with the sunrise and sunset of each day. Therefore, median relative error (MedRE) or the median relative error (MedRE-95%) of day-time forecasting (defined as forecasting when the PV plants are at least operating 5% of their capacities) are the better indicators of relative error alongside MAE as performance metrics.

The MAE, along with MedRE and MedRE-95% are presented against the number of PV plants in the system to evaluate the scalability of the models. The experiments started with the 18 PV plants in West Virginia from the NREL solar power data set. Subsequently, the PV plants from Virginia (221 plants), North Carolina (196 plants), South Carolina (284 plants), Ohio (95 plants), and Georgia (332 plants) are added one at a time to increase system size gradually to assess the scalability of the proposed framework.

4.1. Forecasting in 5–30 minutes horizons with only output power data

For forecasting in 5–30-minute horizons, previous 8 samples of output PV power have been considered as the features. Fig. 6(a)–(b) presents a visual time series illustration of the forecasting performance with an example forecasting results for $t_h = 5$ -minutes and $t_h = 30$ -minutes time horizons at PV plants numbers 7 and 1, respectively. The figures show that the proposed framework is capable of tracking

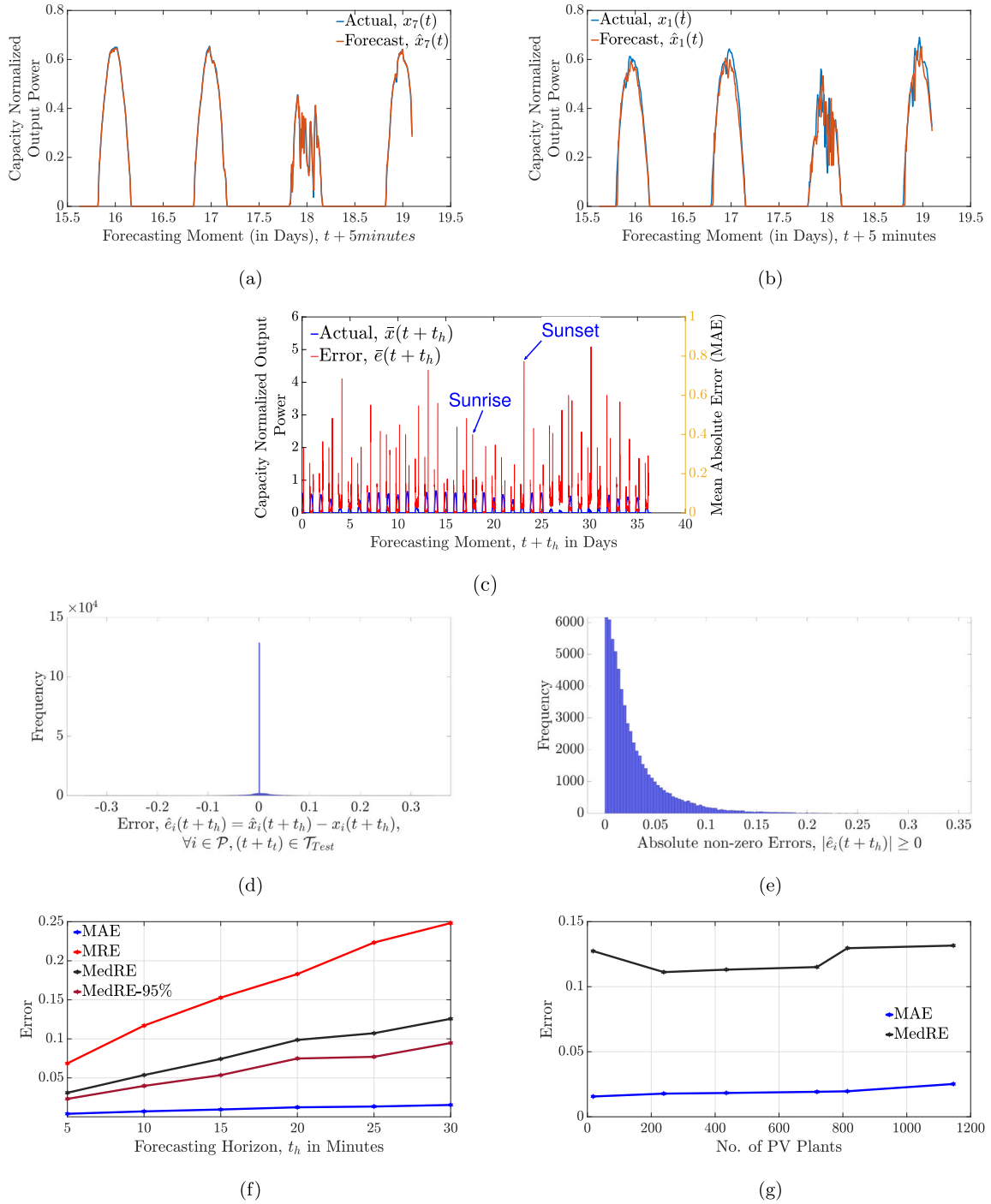
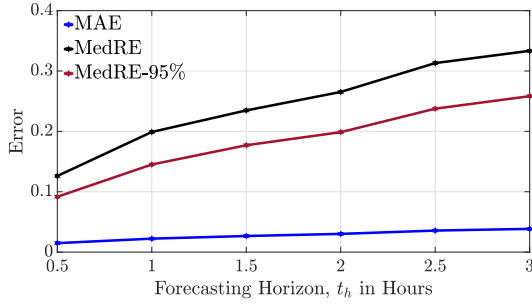


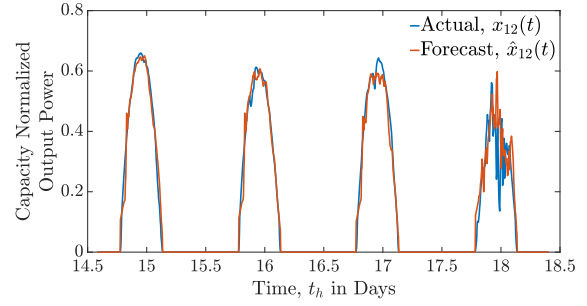
Fig. 6. Forecasting in 5–30 minutes time horizon using only power output sample data: (a)–(b): Example forecast at a sample plant, (a) $t_h = 5$ minutes, (b) $t_h = 30$ minutes, (c) temporal characteristics of absolute error, (d) histogram of errors, (e) histogram of non-zero absolute errors, (f)–(g): The error increases with increasing (f) time horizons, (g) system size i.e., number of plants in the system, P .

the actual PV output time series with precision using only previous output power samples at each plant in this range of time horizon. Fig. 6(c)–(e), illustrate the characteristics of the forecasting errors. Fig. 6(c) reveals that the majority of the large error values are associated with the transition moments, in the vicinity of the sunrise and sunset, shown in terms of the average absolute errors, $\bar{e}(t + t_h)$ calculated over all the plants, where $\bar{e}(t + t_h) = \frac{\sum_{i \in \mathcal{P}} |\hat{e}_i(t + t_h)|}{P}$ and $\bar{x}(t + t_h) = \frac{\sum_{i \in \mathcal{P}} \hat{x}_i(t + t_h)}{P}$. Fig. 6(d) is the histogram of the error values. Both the zero errors during the night-time forecast and a large number of very small errors (≤ 0.005) contribute to the high peak of the histogram. Fig. 6(e) is

the histogram of the non-zero absolute error values, indicating most of the forecasting errors are very small. However, the large tail indicates the presence of a very small number of high error values even in the (≥ 0.2) range. Fig. 6(f) presents the forecasting performance against the forecasting horizon, t_h . The errors increase with increasing t_h , indicating the requirement of other information than the previous output power samples to accurately forecast output power. The scalability of the proposed technique has been assessed in Fig. 6(g) by showing the error against the number of plants in the system. Since all the error metrics exhibit similar increasing and decreasing trends, the rest of the results are presented mostly in terms of only the MAE.



(a) Error vs. forecasting horizon.



(b) Example forecast time series.

Fig. 7. Forecasting in 0.5–3-hr horizon with previous output power samples and ToD index.

4.2. Forecasting in 1–3 hours horizons with output power data and time of the day index

For forecasting within a 30-minute to 3-hour horizon, the past four samples of PV power output, the time of day (ToD) index, and the weather attributes have been selected as input features. Fig. 7(a) illustrates the forecasting performance as a function of the forecasting horizon, $30 \text{ min} \leq t_h \leq 3 \text{ h}$, while Fig. 7(b) shows an example 30-minutes ahead forecasting result in the temporal domain for the plant 12. The results show a similar trend as the performance of the previous range of forecasting horizons. The accuracy for 30-minutes ahead forecasting improves slightly with the inclusion of ToD than the same horizon forecasting with only previous output samples. This justifies $t_h = 30$ -minutes as the appropriate point to introduce other features than the previous output samples. The continuous increase of error with increasing $t_h = 30$ - indicates the demand for weather information longer horizon forecasting beyond 3-hours. The differences in *Med Re* and *Med R-95%* reveal that a significant number of relative errors are due to the very small output power values, especially in the vicinity of sunrise and sunset.

4.3. Day and multiple day ahead forecasting with lower resolution data

For day-ahead and multi-day-ahead forecasting, the previous two samples of PV power output with the time-of-the-day (ToD) index have been chosen as the input features. The weather attributes in the form of the forecast temperature (Temp), total cloud cover (Cloud), and solar parameter canopy conductance (RCS) at the forecasting moment are added to the feature set. Fig. 8(a) demonstrates the forecasting performance to the forecasting horizon t_h , while Fig. 8(b), shows the forecasting accuracy against the number of plants in the system to assess the scalability. It can be observed that the forecasting error for days-ahead forecasting does not necessarily increase with the number of days, i.e., t_h . This is because the weather on two consecutive days can vary significantly, while the weather on one day might closely resemble that of a few days later. This further highlights the greater dependency of forecasting accuracy on weather-related features compared to past output power values, especially when day-to-day weather variations are significant.

The granularity of weather data from NOAA is coarser (1/8th; three hours and five minutes, respectively) than that of the PV output data. One approach to address this mismatch is to linearly interpolate (up-sample and interpolate by a factor of 8) the weather data to align with the resolution of the PV data. However, this resulted in suboptimal outcomes due to inaccuracies in the weather data interpolation, particularly for cloud cover data. As an alternative, the PV data was down-sampled (by a factor of 8) to match the resolution of the weather data for multiple-day predictions. Given the applications of days-ahead forecasting in electric grid scenarios, this granularity adjustment is more acceptable. This approach produced improved results, as shown

in Fig. 8. Additionally, forecasting accuracy can be further enhanced by utilizing more granular weather data when available, requiring only minimal adjustments to the IDPM.

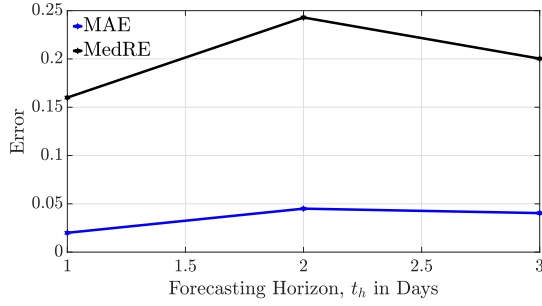
4.4. Comparison

Since the main contributions of the work lie in imparting features and data of various types, sizes, and resolutions into a single framework through the processing in the IDPM module for accommodating generalized-horizon PV power forecasting applicable to different power grid applications, direct comparison with any standalone forecasting technique is not much relevant here. Moreover, particularly, for longer forecasting horizons, where prior processing (such as resolution adjustment) of data-set is needed within the IDPM module, a direct comparison in terms of accuracy and computation time is also not appropriate. Nevertheless, to show the superiority of some of the features over the benchmark techniques, a couple of experiments (up to the forecasting horizon of 30 min) have been discussed in the sections. The persistent method, auto-regressive method, and the Long-short-term-memory (LSTM) have been considered as the benchmark for comparison to assess the accuracy and scalability of the proposed technique. In PV power forecasting literature, the persistent method that assumes the next time step's output to be the same as the current one, i.e., $\hat{x}(t + t_h) = x(t)$ is often used as a benchmark to offer a simple baseline which sometimes performs better than complex tracking methods, especially for shorter horizons [2]. The polynomial auto-regressive model [36,37] tracks the time series by capturing linear temporal correlations using past outputs and is the foundation of many advanced forecasting mechanisms. Finally, the LSTM method, widely regarded as one of the most effective techniques for handling sequence and time-series data, employs a neural network with memory cells to capture long-term dependencies and patterns and is used as the backbone of various state-of-the-art forecasting techniques.

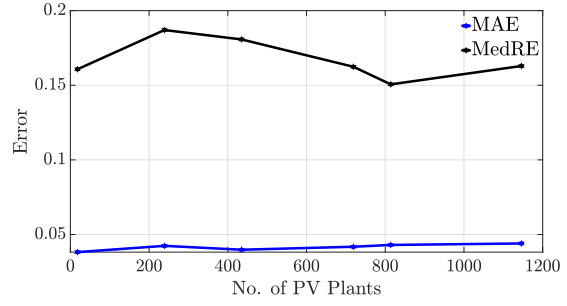
Here, comparisons with the benchmarks are performed on two aspects: the accuracy against the forecasting horizons, in Fig. 9(a) for all West Virginia (18) plants and scalability by observing the 30-minute forecasting accuracy against the number of plants in the system in Fig. 9(b). The comparisons are shown for the 5–30-minute horizons for the consideration of the same sets of features facilitating fair comparison.

Our experiments show that (Fig. 9(a)) the auto-regressive method introduces a very large error (more than 0.13) for all the horizons indicating worse performance than the persistent method itself. Persistent models are appropriate for short-term forecasting, and thereby outperformed by the LSTM method beyond the 15-minute horizon. However, our proposed GAT-based forecasting outperforms all three techniques for all the horizons with only an MAE of 0.0252.

As illustrated in Fig. 9(b), it can be observed that the proposed method is highly scalable. The polynomial auto-regressive techniques fail to converge beyond 25–30 plants and, therefore, are not presented

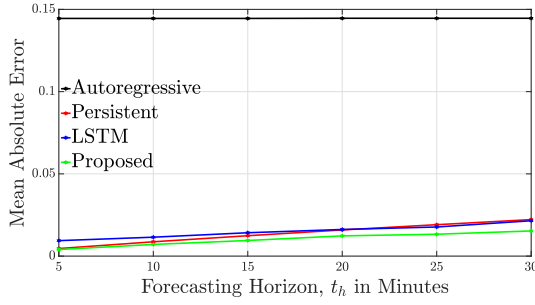


(a) Error vs. forecasting horizon.

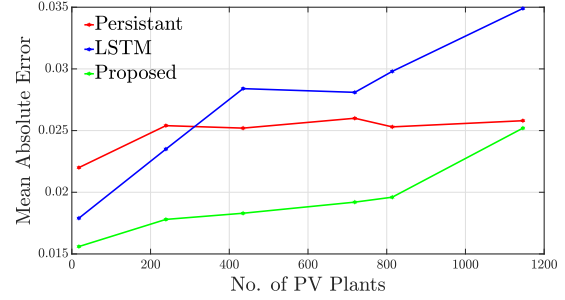


(b) Error vs No. of plants.

Fig. 8. Days-ahead forecasting with 3-hr resolution weather data.



(a) Forecasting accuracy for West Virginia (18) plants in various time horizons.



(b) Scalability comparison.

Fig. 9. Comparison with benchmarks.

in the figure. The proposed technique outperforms LSTM and the persistent method for all numbers of plants in the system while having a less steep MAE vs. system-size (number of plants) curve, indicating better scalability than LSTM. The low steepness of the MAE vs. system-size curve also suggest better scalability of the proposed framework beyond the maximum system-size considered (1046 plants) in the experiments. For very large systems, adjusting graph sparsity can be efficient which is discussed in Section 5.2.

4.5. Computational time

The computational time for training the GAT-based framework mainly depends on the size of the systems (number of plants considered for simultaneous forecasting), the amount of training data, and the number of epochs required to obtain a level of training accuracy. Since these factors vary with the forecasting horizons and features used to forecast the output PV, here the computational time for the three cases discussed above has been reported. However, by keeping the value of L limited, the GAT-based architecture of PV forecasting can ensure scalability. Moreover, by excluding the data with zero output power values at all plants (i.e., associated with night-time output power generation), the number of training data has been reduced without loss of diversity in data. The training time of 50 epochs with 1146 plants together with data from 365-37 = 328 days with only 8 past output samples is 20 min 44 s. For the same number of plants in the 3-hours ahead forecasting with down-sampled data, the training time is 7 min 56 s and for day-ahead forecasting, the training time is 5 min 9 s for 100 epochs each using the set of features described in Table 1. The processing of data within the IDPM module used in this article (resolution adjustment, concatenation, dimension adjustment, etc.) requires 9, 33, and 53 s, respectively, for the three forecasting horizons. In all cases of inferences, with 37 days of test data, the computational time is less than a minute. All the computations are done using an NVIDIA RTX 4090 GPU.

5. Sensitivity analysis and discussions

5.1. Error sensitivity to PV plant type

The forecasting performances of output power in Utility-scale PVs (UPV) and Distributed PVs (DPV) [38] have significant differences as illustrated in Table 2 with three example forecasting experiments: (1) 30-minutes ahead forecasting with 8 previous output power samples, time of the day indices, and baseline weather attributes at the forecasting moment: Temperature, Cloud, and RCS, and (3) day ahead forecasting with 2 previous output power samples, time of the day indices, and baseline weather attributes at the forecasting moment: Temperature, Cloud, and RCS. In all of the three example forecasting cases, the mean absolute error (MAE) is lower for the DPVs. This is because, in DPVs, the high-frequency fluctuations of output power are extensively filtered out compared to the UPVs [32], making the output power time series smooth. Since smooth time series are easier to model and track resulting in smaller forecasting errors.

5.2. Error sensitivity to graph sparsity

The sparsity of the graph is governed by two key parameters: d_i and K as in Eq. (3) discussed detailed in Section 2.2.2. To control graph sparsity, we recommend keeping d_i fixed at 0.0177 and tuning the value of K , as it directly refers to the minimum number of vertices (plants) to which a vertex is connected. Our experiments across various forecasting horizons indicate that $K = 10$ is a stable choice with minimal error deviation. Fig. 10 illustrates the forecasting performance against different values of K , for all PV plants in West Virginia for a fixed, $d_i = 0.0177$. Based on the results, a value of $K = 10$ is recommended, as it provides optimal performance in this experiment and across systems of varying

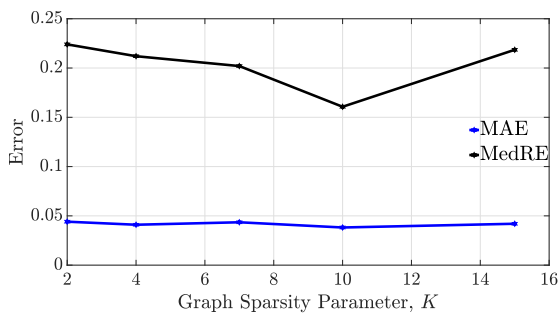
Table 2**Mean absolute error (MAE) by PV type:** Errors are higher utility scale PVs because of the presence of high-frequency fluctuation which are challenging to forecast.

Mean absolute error (MAE)	30-min ahead forecasting (Selected features: 8 previous output samples)	3 h ahead forecasting (Selected features: 4 previous output samples+ToD+weather)	Day ahead forecasting (Selected features: 2 previous output samples+ToD+weather)
Utility-scale PV (UPV)	0.0232	0.0393	0.0474
Distributed PV (DPV)	0.0169	0.0340	0.0407
Combined	0.0200	0.0366	0.0440

Table 3

State-wise mean absolute error (MAE).

Mean absolute error (MAE)	30-min ahead forecasting (Selected features: 8 previous output samples)	3 h ahead forecasting (Selected features: 4 previous output samples+ToD+weather)	Day ahead forecasting (selected features: 2 previous output samples+ToD+weather)
West Virginia	0.0218	0.0364	0.0420
Virginia	0.0197	0.0363	0.0397
North Carolina	0.0192	0.0342	0.0411
South Carolina	0.0183	0.0372	0.0433
Ohio	0.0201	0.0360	0.0375
Georgia	0.0194	0.0379	0.0451

**Fig. 10.** Error sensitivity to graph sparsity parameter K , the minimum number of vertices to which a vertex (plant) would be connected ($d_i = 0.01577$).

sizes. However, in industry implementations, the value K is suggested to be adjusted depending on system sizes, computational resources, and the desired level of accuracy.

5.3. Error sensitivity to plant location

Our experiments for all ranges of forecasting horizons demonstrate no significant variation in average forecasting error with the plant location. This indicates that the model can effectively capture the spatial dynamics related to the weather characteristics and the generation of the PV output power. Table 3 illustrates the state-wise forecasting accuracy in terms of MAE for $t_h = 30$ min, 3 h, 1 day.

5.4. Error sensitivity to seasonality

The sensitivity of the forecasting performance to seasonality has been analyzed by taking various test-training splits, i.e., picking the 10% test data, from various parts of the year 2006. Some slight variation in the average error of forecasting has been observed throughout the year for all three different forecasting horizons (30 min, 3 h, and one day). The general trend observed in the error patterns indicates higher forecasting errors during March and April, and lower errors during November and December. However, it is challenging to draw definitive conclusions from these results based on single-year data. This is because when a portion of data with high variance (e.g., March–April or December–January) is used as test data, the training set lacks diversity due to the exclusion of that portion. Therefore, conducting seasonality analyses using multi-year data would provide more meaningful insights and can be considered as a potential extension of this work.

6. Conclusion and future work

This work introduces a graph attention network-based multi-site PV forecasting framework that supports imparting a variety of data as input depending on the forecasting scenarios and time horizons through an adjustable module, IDPM. This module maps the heterogeneous input data to a fixed-length feature space to facilitate generalized-horizon PV power forecasting with varying sets of weather attributes, which addresses the lack of generalized-horizon frameworks in the PV forecasting literature. The spatiotemporal correlations among the PV plants are then captured by the GAT from the fixed-length features, using a modified geometric distance-based graph. The proposed technique achieves a high forecasting accuracy for various time horizons and is highly scalable and robust for forecasting scenarios. The dependency of forecasting accuracies on particular weather attributes and sensitivity to graph sparsity are presented in the article.

The current work can be extended in several directions. The PV system with the geographical coordinates of the plants can be integrated into an electric grid of suitable size and directly analyze the effect of forecasting error on grid health. In addition, developing advanced forecasting accuracy metrics related to grid stresses can be considered as future work. Moreover, weather data related to the instantaneous cloud movement can be imparted to increase the accuracy of the technique.

CRedit authorship contribution statement

Md Abul Hasnat: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Somayeh Asadi:** Writing – review & editing, Supervision. **Negin Alemazkoo:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Negin Alemazkoo reports financial support was provided by Commonwealth Cyber Initiative. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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